Acquired aplastic anemia

Immune destruction of hematopoietic stem cells · pancytopenia + empty marrow · **grade severity, exclude hypoplastic MDS and inherited BMF, HLA-type early; MSD HCT if young, otherwise hATG + cyclosporine + eltrombopag**

Compiled from ASH 2026, BSH 2024 and ASTCT 2024 guidelines · NIH / EBMT data · FDA labels · primary trial reports verified against PubMed, October 2026.

DEFINITION & SEVERITY

Grade	Criteria
AA	**Pancytopenia + hypocellular marrow** with no abnormal infiltrate, no fibrosis and no MDS-defining dysplasia/ cytogenetics. A diagnosis of exclusion
Severe (SAA) *Camitta*	**Cellularity <25%** (or 25–50% with <30% residual hematopoietic cells) **+ ≥2 of:** ANC <0.5 · plt <20 · retic <60 ×10^9/L
Very severe *Bacigalupo*	SAA criteria **+ ANC <0.2** — highest infection mortality; treat urgently
Non-severe	Hypocellular marrow + cytopenias not meeting SAA. Split by **transfusion dependence**

Severity, age and donor availability drive every treatment decision.

CAUSES

Type	Examples · clues
Idiopathic immune	Most cases. Cytotoxic T cells (IFN-γ, TNF) destroy HSPCs; responds to IST
Drugs / toxins	Chloramphenicol, gold, carbamazepine, felbamate, sulfonamides; **benzene**. Chemo/radiation aplasia is expected, not AA
Hepatitis-associated	AA weeks–months after a **seronegative (non-A–E) hepatitis**, often young men; immune, treat as AA
Other acquired	Pregnancy · **PNH** · **LGL leukemia** · thymoma · SLE · eosinophilic fasciitis · viruses (EBV, HIV); parvovirus B19 = red cells only
Fanconi anemia	Short stature, radial/thumb anomalies, café-au-lait. **Chromosome breakage (DEB/MMC)**. Do not give standard Cy conditioning
Telomere biology	Nail dystrophy, leukoplakia, early graying, **pulmonary fibrosis, cirrhosis**. **Telomere length flow-FISH**; TERT/ TERC. Poor IST response
GATA2	Monocytopenia, B/NK lymphopenia, **MAC/HPV infections**, lymphedema; hypocellular MDS risk
SDS · DBA	SDS: neutropenia + pancreatic insufficiency (**SBDS**). DBA: **red-cell aplasia**, macrocytosis, high eADA

Suspect inherited BMF in the young, with family history, physical stigmata or IST failure — and **screen a sibling donor for the same disorder**.

TREATMENT PATHWAY — SEVERE / VERY SEVERE AA

Matched sibling donor, age ≤40

↓

MSD bone marrow HCT up front
Marrow graft (PBSC worse: more cGVHD, lower OS)
Cy + ATG conditioning; add fludarabine if older
ASH 2026 / BSH: HCT over IST to age 40;
ASTCT: no fixed cutoff for fit older adults

No MSD, age >40, or not fit for HCT

↓

1. hATG + cyclosporine + eltrombopag
hATG 40 mg/kg/d ×4 + CsA from day 1
+ **eltrombopag 150 mg/d ×6 mo** (RACE, NIH)
rATG inferior up front (Scheinberg 2011)

↓

2. Reassess counts at 3 and 6 months
responses are slow — do not judge before 3 mo
responder: CsA continued, then slow taper

↓

Up-front MUD HCT instead
young + severe infection + MUD readily available (BSH); age <40: MUD or IST both acceptable (ASH)

Refractory (no response by 3–6 mo) or relapse

↓

Second line by 6 mo after ATG
<40 y: HCT (MUD or haplo-PTCy) over 2nd ATG
40–60 y: HCT favored · >60 y: HCT or 2nd ATG
relapse: CsA ± EPAG; HCT or 2nd ATG (>60: 2nd ATG)
no HCT: eltrombopag alone (44%, Olnes 2012)

Non-severe AA

↓

By transfusion need
not transfusion-dependent: observe, CBC q1–3 mo
transfusion-dependent or worsening: CsA ± eltrombopag, or treat as SAA (hATG-based IST)

PEARLS

- **HLA-type siblings in week 1** — donor availability is the first branch point
- **Hypocellular marrow in the elderly is normal**; read cellularity for age
- **Never start ATG during active uncontrolled infection** if avoidable; treat infection first
- **Short telomeres or a germline variant change management**: IST works poorly, Fanconi needs reduced-intensity conditioning

WORKUP

- **CBC + diff, retic, smear** — macrocytosis common; **no blasts, no dysplasia**. HbF (raised in inherited BMF)
- **Exclude mimics:** B12, folate, copper; LFTs; **HBV, HCV, HIV, EBV, CMV, parvovirus B19 PCR**; ANA/dsDNA; T-cell flow / TCR clonality (LGL); CT chest if thymoma suspected
- **High-sensitivity PNH flow** (granulocyte FLAER, 0.01%) at diagnosis
- **Marrow aspirate + trephine**: cellularity, dysplasia, blasts, reticulin; **karyotype** (± FISH 5/7) + **myeloid NGS panel**
- **Inherited screen**: DEB/MMC breakage, telomere length flow-FISH, germline panel when suspected
- **HLA-type patient + full siblings at diagnosis**; start an unrelated-donor search early. Ferritin, creatinine, CMV serostatus, pregnancy test

CLONES · AA VS HYPOPLASTIC MDS

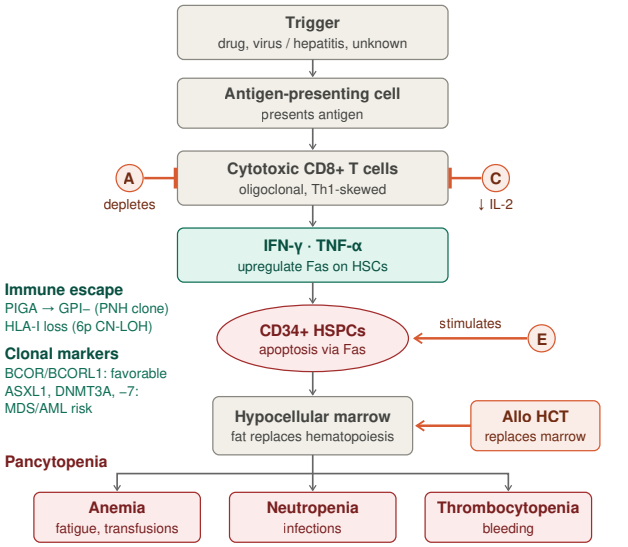
Finding	Meaning
PIGA, BCOR/BCORL1	**Better IST response + survival** *Yoshizato NEJM 2015*
DNMT3A, ASXL1	Clones grow over time; **worse outcome**, higher MDS/AML risk
RUNX1, TP53, –7/7q–	Favor hypoplastic MDS → HCT; –7 after IST = high-risk evolution
Somatic CH	Myeloid-gene mutations in ~1/3, CH in 47%, low VAF — **not MDS by itself**
Morphology	Hypo-MDS: dysplastic megakaryocytes, blasts, CD34+ ↑, fibrosis. AA: fatty marrow, lymphoid aggregates, mild dyserythropoiesis only

SUPPORTIVE CARE

- **Products leukoreduced; irradiated** around ATG and for HCT candidates. **No family-donor transfusions before HCT** — sensitizes to donor antigens, raises graft rejection
- **Prophylactic platelets at <10** (higher if febrile/bleeding); restrictive RBC thresholds
- **ANC <0.5: mold-active azole + antibacterial prophylaxis** *ASH 2026*; HSV/VZV prophylaxis with ATG. **Azoles raise CsA levels**
- **No routine G-CSF** — no survival benefit (Tichelli 2011); short-term use only in severe infection
- **Iron overload: chelate** (deferasirox or deferoxamine); **avoid deferiprone** (agranulocytosis). Phlebotomy after recovery

Acquired aplastic anemia

PATHOPHYSIOLOGY



A horse ATG: depletes T cells · C cyclosporine: calcineurin inhibitor · E eltrombopag: TPO-receptor (MPL) agonist · HCT allogeneic HCT: new donor marrow

IST (horse ATG + cyclosporine) removes the T-cell attack; eltrombopag expands surviving HSCs. Small PNH clones mark immune AA; ASXL1/DNMT3A/–7 clones flag later MDS/AML.

RESPONSE & TIMING — NIH CRITERIA

Response	Definition
Complete	**ANC ≥1.0, plt ≥100, Hb ≥10** (all three)
Partial	**No longer meets SAA criteria**; usually transfusion-independent
None	Still meets SAA criteria

Assess at **3 and 6 months**. Median time to first response **3.0 mo with eltrombopag vs 8.8 mo without** (RACE). Start second-line therapy **no later than 6 mo after ATG** (ASH 2026).

RELAPSE & CLONAL EVOLUTION

- **Relapse 39% at 4 yr** in responders on CsA maintenance — clustered after the 6-mo CsA cut / eltrombopag stop and after CsA stops at 2 yr; **~2/3 respond to CsA ± eltrombopag again** *Patel Blood 2022, n=178*
- **Clonal evolution 15% at 4 yr**; high-risk (myeloid neoplasm or chr 7) **5.7%**, with worse OS — not higher than IST alone, but earlier
- **New cytopenia or falling counts → repeat marrow + karyotype**; –7 or MDS/AML → HCT
- **PNH clones:** small clones are common in AA and predict IST response; they can expand after IST into hemolytic PNH — **re-check flow at least yearly**; complement inhibition only for clinically significant hemolysis

AGENTS — DOSING

Agent	Dosing	Notes
Horse ATG *Atgam*	**40 mg/kg/d IV ×4 days** over ~4 h (NIH; label: 10–20 mg/kg/d ×8–14 d). Premed acetaminophen + antihistamine; **prednisone 1 mg/kg/d ×10 d, taper over 7–14 d**	Infusion reactions; **serum sickness** (fever, rash, arthralgia) ~day 7–14. Platelets before each dose; inpatient
Rabbit ATG *Thymoglobulin*	Second IST course or HCT conditioning	Inferior to hATG up front
Cyclosporine	**3 mg/kg PO q12h** (6 mg/kg/d) from day 1 → **trough 200–400 µg/L**. Responders: continue, then taper slowly (NIH: 2 mg/kg/d to 24 mo)	Nephrotoxicity, HTN, Mg wasting, gingival hyperplasia, PRES. **Azole interactions**
Eltrombopag *Promacta*	**First-line: 150 mg daily ×6 mo** with IST (≥12 y; label ≥2 y, weight/age-based); **75 mg if East/Southeast Asian ancestry** or hepatic impairment. **Refractory: 50 mg → max 150 mg**; stop if no response by 16 wk	**Take ≥2 h before or 4 h after cations** (antacids, calcium, iron, dairy). LFTs; cataract exam; –7 in 4 (D1–M6 cohort), 7/153 across cohorts. FDA: refractory 2014, first-line 2018
Romiplostim	Japan-approved (refractory AA 2019, all AA 2023); **not FDA-approved for AA**	Asian phase 2/3 data only

ALLOGENEIC HCT

- **MSD vs URD** (EBMT, n=1448): URD survival not significantly worse in multivariate analysis, but **more acute (25% vs 13%) and chronic GVHD (26% vs 14%)**. Worst predictor: **PBSC graft**; also ≥180 d from diagnosis, age ≥20, no ATG *Bacigalupo 2015*
- **Marrow over PBSC**: age <20 MSD, 5-yr OS **85% vs 73%** *Schrezenmeier Blood 2007*
- **Haplo with PTCy up front** (Hopkins phase 2, n=27): **3-yr OS 92%**; aGVHD II–IV 7%, cGVHD 4%; with 400 cGy TBI **20/20 alive** *DeZern Blood 2023*
- **Guidelines differ:** ASTCT 2024 favors MUD/haplo over IST when no MSD; ASH 2026 still favors **IST over haplo up front at any age**. Conditioning: rATG, fludarabine for MUD/haplo

KEY TRIALS

Trial	Result → practice
hATG vs rATG *NEJM 2011*	Randomized, n=120: 6-mo response **68% vs 37%**; 3-yr OS 96% vs 76%. **Horse ATG is standard**
NIH IST + EPAG *NEJM 2017*	Phase 1–2, n=92: eltrombopag day 1–6 mo cohort **CR 58%, ORR 94%** vs historical CR 10%, ORR 66%
RACE *NEJM 2022*	Phase 3, n=197, hATG + CsA ± EPAG: **CR at 3 mo 22% vs 10%** (OR 3.2); ORR 6 mo 68% vs 41%; no extra toxicity
Refractory EPAG *NEJM 2012*	Single-arm, n=25: **44% response** at 12 wk, multilineage
EBMT G-CSF *Blood 2011*	Randomized, n=192: fewer infections, **no OS/EFS benefit** — no routine G-CSF
Hopkins haplo *Blood 2023*	Up-front haplo-PTCy, n=27: 3-yr OS 92%, minimal GVHD

PREGNANCY

- AA can present or relapse in pregnancy; after IST, **19% of 36 pregnancies relapsed**, 34 live births *Tichelli 2002*
- **Supportive care** (transfusion) is the mainstay; **CsA can be continued**; avoid ATG and eltrombopag
- Low platelets and a PNH clone predict complications — consider LMWH if a PNH clone is present